LIBRARY DATABASES

1. Create Operation:

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\parja>mongosh
Current Mongosh Log ID: 69f49f296ba62fd21c3682d0
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Using MongoDB:          8.2.7
Using Mongosh:          2.8.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-04-30T18:32:11.440+05:30: Access control is not enabled for the database. Read and write access to data and configuration is
unrestricted
------

test> show databases
admin      40.00 KiB
config    108.00 KiB
libraray    8.00 KiB
local      40.00 KiB
school     68.00 KiB
test> use library
switched to db library
library> db.book.find()

library> db.book.drop()
true
library>
```

2. Insert data:

```
library> db.books.insertMany([
|    {
|      book_id: 1,
|      title: 'The Alchemist',
|      author: 'Paulo Coelho',
|      genre: 'Fiction',
|      published_year: 1988,
|      available_copies: 5
|    },
|    {
|      book_id: 2,
|      title: 'Clean Code',
|      author: 'Robert C. Martin',
|      genre: 'Programming',
|      published_year: 2008,
|      available_copies: 3
|    },
|    {
|      book_id: 3,
|      title: 'Introduction to Algorithms',
|      author: 'Thomas H. Cormen',
|      genre: 'Education',
|      published_year: 2009,
|      available_copies: 4
|    },
|    {
|      book_id: 4,
|      title: "Harry Potter and the Sorcerer's Stone",
|      author: 'J.K. Rowling',
|      genre: 'Fantasy',
|      published_year: 1997,
|      available_copies: 6
|    },
|    {
|      book_id: 5,
|      title: 'The Pragmatic Programmer',
|      author: 'Andrew Hunt',
|      genre: 'Programming',
|      published_year: 1999,
|      available_copies: 2
|    }
| ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f4a7176ba62fd21c3682da'),
    '1': ObjectId('69f4a7176ba62fd21c3682db'),
    '2': ObjectId('69f4a7176ba62fd21c3682dc'),
    '3': ObjectId('69f4a7176ba62fd21c3682dd'),
    '4': ObjectId('69f4a7176ba62fd21c3682de')
  }
}
```

3. **Delete data:**

```
library> db.books.find()
[
  {
    _id: ObjectId('69f4a1186ba62fd21c3682d1'),
    id: 1,
    name: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a1586ba62fd21c3682d2'),
    id: 1,
    name: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  }
]
library> db.books.deleteOne({id: 1})
{ acknowledged: true, deletedCount: 1 }
library> db.books.find()
[
  {
    _id: ObjectId('69f4a1586ba62fd21c3682d2'),
    id: 1,
    name: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
```

4. **Read data:**

```
library> db.books.find()
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 2
  }
]
```

5. Update data:

```
library> db.books.updateOne(
|    { book_id: 5 },
|    { $set: { available_copies: 7 } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.find()
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
```

6. Greater than and Lesser Than Operation:

```
library> db.books.find({ available_copies: { $gt: 5 } });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
]
library> db.books.find({ available_copies: { $lt: 5 } });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  }
]
```

7. **In Operation:**

```
library> db.books.find({ available_copies: { $in: [1, 3, 4] } });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  }
]
```

8. **Or Operation:**

```
library> db.books.find({
| $or: [
| {genre: "Fantasy"},
| {published_year: {$lt: 2000} }
| ]
| });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
]
```

9. And Operation:

```
library> db.books.find({
| $and: [
| {genre: "programming"},{
| published_year: 2008}]
| });
|

library> db.books.find({
| $and: [
| {genre: "Programming"},
| {available_copies: {$gt: 2} }
| ]
| });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
]
library> db.books.find({
| genre: "Programming",
| available_copies: {$gt: 2}
| });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
]
```

10. Exists:

```
library> db.books.find({ book_id: { $exists: true } });
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
]
```

11. Limit and Skip Opeation:

```
library> db.books.find().limit(2);
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  }
]
library> db.books.find().skip(4);
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  }
]
```

12. Ascending Operation:

```
library> db.books.find().sort({published_year: 1});
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  }
]
```

13. Descending Operation;

```
library> db.books.find().sort({published_year: -1});
[
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dc'),
    book_id: 3,
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    genre: 'Education',
    published_year: 2009,
    available_copies: 4
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682db'),
    book_id: 2,
    title: 'Clean Code',
    author: 'Robert C. Martin',
    genre: 'Programming',
    published_year: 2008,
    available_copies: 3
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682de'),
    book_id: 5,
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    genre: 'Programming',
    published_year: 1999,
    available_copies: 7
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682dd'),
    book_id: 4,
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    genre: 'Fantasy',
    published_year: 1997,
    available_copies: 6
  },
  {
    _id: ObjectId('69f4a7176ba62fd21c3682da'),
    book_id: 1,
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    genre: 'Fiction',
    published_year: 1988,
    available_copies: 5
  }
]
```